12TH CHEMISTRY · CHAPTER 24 · FULL CHAPTER MCQS

Organic Nitrogen Compounds

122 topic-wise MCQs from the complete chapter with answers and reasons

INTRODUCTION AND 24.1 AMINES

1. Proteins are built from:

(a) Nucleotides (b) Amino acids
(c) Fatty acids (d) Esters

Answer: (b) Amino acids — Organic nitrogen compounds form the basis of life.

2. Which is a nitrogen-containing polymer?

(a) Polythene (b) Polystyrene
(c) PVC (d) Nylon

Answer: (d) Nylon — Also Kevlar and polyurethane.

3. The nitrogen compound present in coffee is:

(a) Theobromine (b) Aniline
(c) Caffeine (d) Glycine

Answer: (c) Caffeine — Theobromine is in chocolate.

4. Theobromine is found in:

(a) Chocolate (b) Coffee
(c) Vinegar (d) Fish

Answer: (a) Chocolate — Stated in the textbook.

5. Amines are organic derivatives of:

(a) Water (b) Methane
(c) Nitric acid (d) Ammonia

Answer: (d) Ammonia — Hydrogen atoms of ammonia are replaced by alkyl or aryl groups.

6. In the IUPAC system amines are named as:

(a) Alkanols (b) Alkanamides
(c) Alkanamines (d) Alkanenitriles

Answer: (c) Alkanamines — E.g. methanamine.

7. A primary amine contains the group:

(a) $-NH_2$ (b) >NH
(c) >N– (d) $-NO_2$

Answer: (a) $-NH_2$ — One alkyl or aryl group on nitrogen.

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8. N-Methylethanamine is a:

(a) Primary amine (b) Secondary amine
(c) Tertiary amine (d) Quaternary salt

Answer: (b) Secondary amine — Two alkyl groups on nitrogen.

9. N,N-Dimethylmethanamine is a:

(a) Primary amine (b) Secondary amine
(c) Tertiary amine (d) Amide

Answer: (c) Tertiary amine — Three methyl groups on nitrogen.

10. Phenylamine is also called:

(a) Aniline (b) Anisole
(c) Toluene (d) Phenol

Answer: (a) Aniline — It is a primary aromatic amine.

11. Diphenylamine is a:

(a) Primary amine (b) Secondary amine
(c) Tertiary amine (d) Diazo compound

Answer: (b) Secondary amine — Two aryl groups on the NH group.

12. Triphenylamine is a:

(a) Primary amine (b) Quaternary salt
(c) Secondary amine (d) Tertiary amine

Answer: (d) Tertiary amine — Three aryl groups on nitrogen.

13. Tetramethylammonium chloride is a:

(a) Tertiary amine (b) Amide
(c) Quaternary ammonium salt (d) Nitrile

Answer: (c) Quaternary ammonium salt — Tetra-substituted nitrogen.

14. The class of an amine is decided by the branching at the:

(a) Carbon attached to nitrogen (b) Benzene ring
(c) Last carbon (d) Nitrogen atom

Answer: (d) Nitrogen atom — Unlike alcohols and alkyl halides.

15. The IUPAC name of CH_3–CH_2–CH_2–NH_2 is:

(a) Propan-1-amine (b) Ethanamine
(c) Propan-2-amine (d) N-Methylethanamine

Answer: (a) Propan-1-amine — Three carbon atoms, –NH2 on C-1.

16. The number of isomeric amines of formula C_3H_9N is:

(a) Two (b) Four
(c) Three (d) Five

Answer: (b) Four — Two primary, one secondary, one tertiary.

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24.2 PREPARATION OF AMINES

17. Bromoethane with excess ethanolic ammonia under pressure gives:

(a) Ethanamide
(b) Ethanenitrile
(c) Ethanamine
(d) Ethanol

Answer: (c) Ethanamine — Nucleophilic substitution.

18. Excess ammonia is used with a halogenoalkane to avoid formation of:

(a) Alcohols
(b) Secondary amines or a mixture
(c) Alkenes
(d) Nitriles

Answer: (b) Secondary amines or a mixture — The primary amine is then the major product.

19. If the halogenoalkane is in excess, the major product is a:

(a) Primary amine
(b) Amide
(c) Nitrile
(d) Tertiary amine

Answer: (d) Tertiary amine — Stated in the textbook box.

20. Reduction of propanenitrile gives:

(a) Propan-1-amine
(b) Ethanamine
(c) Propan-2-amine
(d) Propanamide

Answer: (a) Propan-1-amine — With $LiAlH_4$ or H_2/Ni.

21. Nitriles are reduced to primary amines by:

(a) $K_2Cr_2O_7$
(b) Br_2/NaOH
(c) $NaNO_2$/HCl
(d) H_2/Ni or $LiAlH_4$

Answer: (d) H_2/Ni or $LiAlH_4$ — Given in the textbook.

22. Reduction of nitroethane gives:

(a) Ethanamide
(b) Ethanol
(c) Ethanamine
(d) Ethanenitrile

Answer: (c) Ethanamine — A primary amine and water.

23. Reduction of ethanamide with lithium tetrahydridoaluminate gives:

(a) Ethanamine
(b) Methanamine
(c) Ethanol
(d) Ethanoic acid

Answer: (a) Ethanamine — No by-product, so the yield is good.

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24. The Hofmann reaction of ethanamide gives:

(a) Ethanamine (b) Methanamine
(c) Propanamine (d) Ethanenitrile

Answer: (b) Methanamine — One carbon less than the amide.

25. The reagents of the Hofmann reaction are:

(a) Sn and HCl (b) $LiAlH_4$ and ether
(c) $NaNO_2$ and HCl (d) Br_2 and NaOH

Answer: (d) Br_2 and NaOH — Applied to an amide.

26. The amine from the Hofmann reaction has, compared with the amide:

(a) One carbon more (b) The same carbons
(c) One carbon less (d) Two carbons less

Answer: (c) One carbon less — Carbon is lost as Na_2CO_3.

27. Bromoethane with methanamine in ethanol gives:

(a) N-Methylethanamine
(b) Propan-1-amine
(c) Ethanamine
(d) N,N-Dimethylethanamine

Answer: (a) N-Methylethanamine — Alkylation of a primary amine gives a secondary amine.

28. Reduction of N-methylethanamide gives:

(a) Ethanamine (b) N-Methylethanamine
(c) Methanamine (d) Propan-1-amine

Answer: (b) N-Methylethanamine — N-substituted amides give secondary amines.

29. Phenylamine is prepared from chlorobenzene by:

(a) Hydrolysis (b) Diazotization
(c) Ammonolysis (d) Coupling

Answer: (c) Ammonolysis — With NH_3, Cu_2O, 250 °C, 50 atm.

30. The catalyst in the ammonolysis of chlorobenzene is:

(a) Cu_2O (b) Ni
(c) Sn (d) $AlCl_3$

Answer: (a) Cu_2O — At 250 °C and 50 atm.

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31. Nitrobenzene is reduced to phenylamine by:

(a) $NaNO_2$ / HCl
(b) $KMnO_4$ / H_2SO_4
(c) Br_2 / NaOH
(d) Sn / conc. HCl

Answer: (d) Sn / conc. HCl — 6[H] are needed.

32. The Hofmann reaction of benzamide gives:

(a) Benzylamine (b) Phenylamine
(c) Benzoic acid (d) Benzonitrile

Answer: (b) Phenylamine — One carbon less.

33. 2-Phenylethanamine is known as the:

(a) Fish odour compound
(b) Butter yellow dye
(c) “Feel good factor” of chocolate
(d) Locating agent

Answer: (c) “Feel good factor” of chocolate — Quick Check 24.2.

24.3 BASICITY AND REACTIONS OF AMINES

34. Aqueous solutions of amines are:

(a) Strongly acidic (b) Strongly alkaline
(c) Neutral (d) Weakly alkaline

Answer: (d) Weakly alkaline — They partly react with water to give OH^- ions.

35. The basicity of an amine depends on the availability of the:

(a) Hydrogen atoms (b) Alkyl carbon
(c) Lone pair on nitrogen (d) π electrons

Answer: (c) Lone pair on nitrogen — It accepts a proton.

36. A stronger base has a:

(a) Smaller K_b (b) Smaller pK_b
(c) Larger pK_b (d) Zero K_b

Answer: (b) Smaller pK_b — Larger K_b means smaller pK_b.

37. The pK_b of ammonia is:

(a) 4.75 (b) 3.29
(c) 9.38 (d) 14

Answer: (a) 4.75 — Table 24.1.

38. The pK_b of ethanamine is:

(a) 4.75 (b) 9.38
(c) 3.29 (d) 10

Answer: (c) 3.29 — Table 24.1.

39. The pK_b of phenylamine is:

(a) 3.29 (b) 9.38
(c) 4.75 (d) 12

Answer: (b) 9.38 — Table 24.1.

40. The K_b of ammonia is:

(a) 1.8×10^{-5} (b) 5.1×10^{-4}
(c) 4.2×10^{-10} (d) 1.0×10^{-14}

Answer: (a) 1.8×10^{-5} — Table 24.1.

41. The correct order of basic strength is:

(a) Phenylamine > ammonia > ethanamine
(b) Ammonia > phenylamine > ethanamine
(c) Ammonia > ethanamine > phenylamine
(d) Ethanamine > ammonia > phenylamine

Answer: (d) Ethanamine > ammonia > phenylamine — From the pK_b values.

42. Ethanamine is a stronger base than ammonia because the ethyl group is:

(a) Electron-withdrawing (b) Electron-donating
(c) Acidic (d) Bulky

Answer: (b) Electron-donating — It makes the lone pair more available.

43. Phenylamine is a weak base because its lone pair is:

(a) Absent
(b) Bonded to hydrogen
(c) Delocalised over the benzene ring
(d) Localised on nitrogen

Answer: (c) Delocalised over the benzene ring — It overlaps with the π electrons of the ring.

44. In phenylamine the lone pair of nitrogen lies in a:

(a) s orbital (b) σ bond
(c) d orbital (d) p orbital

Answer: (d) p orbital — The nitrogen atom is planar.

45. Amines react with acids to form:

(a) Salts (b) Amides
(c) Nitriles (d) Esters

Answer: (a) Salts — Nitrogen forms a dative bond with a proton.

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46. Ethanamine with HCl gives:

(a) Ethanamide (b) Chloroethane
(c) Ethylammonium chloride (d) Ethanenitrile

Answer: (c) Ethylammonium chloride — $CH_3CH_2NH_3^+Cl^-$.

47. The foul smell of fish is due to:

(a) N,N-Dimethylmethanamine (b) Ethanamine
(c) Phenylamine (d) Ethanamide

Answer: (a) N,N-Dimethylmethanamine — Trimethylamine.

48. The acid in lemon juice that reacts with fish amines is:

(a) Ethanoic acid (b) Citric acid
(c) Tartaric acid (d) Methanoic acid

Answer: (b) Citric acid — It makes the fish less smelly.

49. Amines are distinguished from one another by reaction with:

(a) Nitric(V) acid (b) Ethanoic acid
(c) Sulphuric acid (d) Nitrous acid

Answer: (d) Nitrous acid — Nitric(III) acid.

50. Primary amines with nitrous acid form:

(a) N-Nitrosoamines (b) Diazonium salts
(c) Nitrite salts (d) Amides

Answer: (b) Diazonium salts — The process is diazotization.

51. Secondary amines with nitrous acid give:

(a) A yellow oily liquid (b) Nitrogen gas
(c) A soluble salt (d) A white precipitate

Answer: (a) A yellow oily liquid — N-nitrosoamines.

52. Tertiary amines with nitrous acid give:

(a) Nitrogen gas (b) A yellow oil
(c) A soluble nitrite salt (d) An azo dye

Answer: (c) A soluble nitrite salt — No visible sign of reaction.

53. The observation when a primary amine reacts with nitrous acid is:

(a) Yellow oil (b) White precipitate
(c) No change (d) Bubbles of nitrogen gas

Answer: (d) Bubbles of nitrogen gas — Summary in the textbook.

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54. Nitrous acid is prepared in situ from:

(a) $NaNO_2$ and HCl
(b) $NaNO_3$ and HCl
(c) HNO_3 and H_2SO_4
(d) NH_3 and HCl

Answer: (a) $NaNO_2$ and HCl — It is unstable.

55. Cyclohexylamine is more basic than phenylamine by a factor of about:

(a) Ten
(b) Two
(c) A hundred
(d) A million

Answer: (d) A million — Quick Check 24.3.

24.3.3 PHENYLAMINE AND AZO COMPOUNDS

56. The amino group in phenylamine is:

(a) Meta directing
(b) Ortho-para directing
(c) Deactivating
(d) Non-directing

Answer: (b) Ortho-para directing — It makes the ring fairly reactive.

57. Phenylamine with bromine water gives:

(a) 4-Bromophenylamine
(b) 3-Bromophenylamine
(c) Bromobenzene
(d) 2,4,6-Tribromophenylamine

Answer: (d) 2,4,6-Tribromophenylamine — A white precipitate.

58. Moles of bromine that react with one mole of phenylamine:

(a) One
(b) Two
(c) Three
(d) Six

Answer: (c) Three — Two ortho positions and one para position.

59. The nitrogen atom of the amino group in phenylamine is described in the textbook as:

(a) sp^3 hybridized
(b) sp hybridized
(c) dsp^2 hybridized
(d) Unhybridized

Answer: (a) sp^3 hybridized — The ring carbons are sp^2 hybridized.

60. The diazonium group is:

(a) –N=N–
(b) $–NO_2$
(c) $–N^+{\equiv}N$
(d) $–NH_2$

Answer: (c) $–N^+{\equiv}N$ — Compounds with it are diazo compounds.

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61. Diazotization of phenylamine is carried out:

(a) Above 60 °C
(b) At 250 °C
(c) At 100 °C
(d) Below 10 °C

Answer: (d) Below 10 °C — Diazonium salts decompose easily.

62. Diazonium salts decompose giving off:

(a) Oxygen
(b) Nitrogen
(c) Ammonia
(d) Hydrogen

Answer: (b) Nitrogen — Even at room temperature.

63. Benzenediazonium chloride heated with water gives:

(a) Phenol
(b) Benzene
(c) Phenylamine
(d) Chlorobenzene

Answer: (a) Phenol — With N_2 and HCl.

64. Alkyl diazonium salts are:

(a) Stable at room temperature
(b) Coloured dyes
(c) Highly unstable under all conditions
(d) Stable when dry

Answer: (c) Highly unstable under all conditions — They cannot be isolated.

65. Aryl diazonium salts are more stable because the positive charge is:

(a) Localised on nitrogen
(b) Delocalised over the aromatic ring
(c) Removed by water
(d) On chlorine

Answer: (b) Delocalised over the aromatic ring — Stated in the textbook.

66. The reaction of a diazonium salt with phenol is called:

(a) Diazotization
(b) Hydrolysis
(c) Ammonolysis
(d) Coupling

Answer: (d) Coupling — It gives an azo compound.

67. In coupling, the diazonium ion acts as:

(a) An electrophile
(b) A nucleophile
(c) A base
(d) A reducing agent

Answer: (a) An electrophile — It is positively charged.

68. The diazonium ion attacks phenol at the:

(a) Ortho position only
(b) Para position
(c) Meta position
(d) Oxygen atom

Answer: (b) Para position — Stated in the textbook.

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69. p-Hydroxyazobenzene is:

(a) A white solid
(b) A colourless gas
(c) A blue liquid
(d) An orange solid

Answer: (d) An orange solid — It is an azo compound.

70. Coupling with phenol takes place in:

(a) Mild alkaline solution
(b) Strong acid
(c) Neutral ether
(d) Concentrated acid

Answer: (a) Mild alkaline solution — With aromatic amines: mild acidic conditions.

71. Butter yellow is:

(a) Methyl orange
(b) p-Hydroxyazobenzene
(c) p-Dimethylaminoazobenzene
(d) 2,4,6-Tribromophenylamine

Answer: (c) p-Dimethylaminoazobenzene — It was used in food.

72. Butter yellow is now forbidden in food because it is:

(a) Colourless
(b) Insoluble
(c) Expensive
(d) Carcinogenic

Answer: (d) Carcinogenic — Stated in the textbook.

73. Methyl orange is used as:

(a) An acid–base indicator
(b) A food colour
(c) An antiseptic
(d) A fuel

Answer: (a) An acid–base indicator — It is an azo dye.

74. Methyl orange is prepared from N,N-dimethylphenylamine and diazotized:

(a) Phenylamine
(b) Sulfanilic acid
(c) Phenol
(d) Benzoic acid

Answer: (b) Sulfanilic acid — Stated in the textbook.

75. The azo group in azo dyes is called a:

(a) Auxochrome
(b) Zwitterion
(c) Chromophore
(d) Peptide link

Answer: (c) Chromophore — It is responsible for the intense colour.

24.4 ALKANENITRILES AND 24.5 AMIDES

76. The functional group of alkanenitriles is:

(a) $-NH_2$
(b) –CN
(c) $-NO_2$
(d) $-CONH_2$

Answer: (b) –CN — E.g. ethanenitrile.

77. CH_3–CH_2–CN is named:

(a) Propanenitrile (b) Ethanenitrile
(c) Ethanamine (d) Propanamide

Answer: (a) Propanenitrile — The carbon of –CN is counted.

78. Acid hydrolysis of ethanenitrile gives:

(a) Ethanamide (b) Ethanol
(c) Ethanoic acid (d) Ethanamine

Answer: (c) Ethanoic acid — With ammonium chloride.

79. Alkaline hydrolysis of ethanenitrile gives sodium ethanoate and:

(a) N_2 (b) H_2
(c) NH_4Cl (d) NH_3

Answer: (d) NH_3 — The free acid is obtained on acidification.

80. The functional group of amides is:

(a) $–CONH_2$ (b) –COOH
(c) –COCl (d) –CN

Answer: (a) $–CONH_2$ — They are also called acid amides.

81. Ethanoyl chloride with ammonia gives:

(a) Ethanamine (b) Ethanenitrile
(c) Ethanamide (d) Ethanoic acid

Answer: (c) Ethanamide — At room temperature.

82. Benzoyl chloride with ammonia gives:

(a) Phenylamine (b) Benzamide
(c) Benzoic acid (d) Benzonitrile

Answer: (b) Benzamide — With HCl.

83. Ethanoyl chloride with ethanamine gives:

(a) Ethanamide (b) Diethylamine
(c) N-Methylethanamide (d) N-Ethylethanamide

Answer: (d) N-Ethylethanamide — An N-substituted amide.

84. Ethanoyl chloride with phenylamine gives:

(a) Benzamide (b) N-Phenylethanamide
(c) Phenyl ethanoate (d) Azobenzene

Answer: (b) N-Phenylethanamide — An N-substituted amide.

85. The white fumes released when an acyl chloride reacts with ammonia are of:

(a) Ammonia (b) Chlorine
(c) Hydrogen chloride (d) Water

Answer: (c) Hydrogen chloride — It then forms a salt with excess ammonia.

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86. The only amide that is a liquid is:

(a) Methanamide (b) Ethanamide
(c) Benzamide (d) Propanamide

Answer: (a) Methanamide — All others are solids.

87. The high boiling points of amides are due to:

(a) Ionic bonding (b) Resonance
(c) Large size (d) Hydrogen bonding

Answer: (d) Hydrogen bonding — Strong intermolecular hydrogen bonds.

88. Aqueous solutions of amides are:

(a) Alkaline (b) Neutral
(c) Acidic (d) Strongly basic

Answer: (b) Neutral — Unlike amines.

89. Amides are weak bases because the lone pair on nitrogen is:

(a) Delocalised with the carbonyl oxygen
(b) Absent
(c) Used in hydrogen bonding
(d) On carbon

Answer: (a) Delocalised with the carbonyl oxygen — Resonance.

90. The C–N bond length in an amide is:

(a) 147 pm (b) 120 pm
(c) 154 pm (d) 132 pm

Answer: (d) 132 pm — It has double-bond character.

91. The C–N bond length in an amine is:

(a) 132 pm (b) 120 pm
(c) 147 pm (d) 110 pm

Answer: (c) 147 pm — A normal single bond.

92. The least reactive of all acid derivatives are:

(a) Acyl chlorides (b) Amides
(c) Esters (d) Anhydrides

Answer: (b) Amides — The carbonyl carbon is less electrophilic.

93. Acid hydrolysis of ethanamide gives ethanoic acid and:

(a) NH_3 (b) NaCl
(c) N_2 (d) NH_4Cl

Answer: (d) NH_4Cl — With HCl.

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94. Hydrolysis of ethanamide with NaOH gives sodium ethanoate and:

(a) NH_4Cl (b) N_2
(c) NH_3 (d) H_2

Answer: (c) NH_3 — Base-catalysed hydrolysis.

95. Reduction of benzamide with lithium tetrahydridoaluminate gives:

(a) $C_6H_5–CH_2–NH_2$
(b) Phenylamine
(c) Benzoic acid
(d) Benzene

Answer: (a) $C_6H_5–CH_2–NH_2$ — A primary amine.

96. Amides are reduced in:

(a) Water (b) Ethanol
(c) Dry ether (d) Dilute acid

Answer: (c) Dry ether — Dry ethoxyethane.

24.6 AMINO ACIDS AND PEPTIDES

97. Amino acids contain the groups:

(a) $–NH_2$ and –OH (b) $–NO_2$ and –COOH
(c) $–NH_2$ and –COOH (d) –CN and –OH

Answer: (c) $–NH_2$ and –COOH — A basic and an acidic group.

98. Amino acids are amphoteric because they:

(a) Are coloured (b) Are insoluble
(c) Are neutral (d) Behave as both acid and base

Answer: (d) Behave as both acid and base — They have both groups.

99. The simplest amino acid is:

(a) Glycine (b) Alanine
(c) Lysine (d) Leucine

Answer: (a) Glycine — R is a hydrogen atom.

100. The R group of alanine is:

(a) –H (b) $–CH_3$
(c) $–CH_2OH$ (d) $–C_6H_5$

Answer: (b) $–CH_3$ — 2-Aminopropanoic acid.

101. Glycine is:

(a) 2-Aminopropanoic acid (b) Aminomethanoic acid
(c) Aminoethanoic acid (d) 2-Aminobutanoic acid

Answer: (c) Aminoethanoic acid — The simplest amino acid.

102. The amino acid which is not chiral is:

(a) Glycine (b) Alanine
(c) Lysine (d) Aspartic acid

Answer: (a) Glycine — All others are chiral.

103. Out of 20 α-amino acids, the number of essential ones is:

(a) 5 (b) 20
(c) 15 (d) 10

Answer: (d) 10 — The body cannot synthesise them.

104. A zwitterion is a:

(a) Free radical (b) Dipolar ion
(c) Carbocation (d) Carbanion

Answer: (b) Dipolar ion — It has a positive and a negative end.

105. In the formation of a zwitterion the proton moves from:

(a) Carboxyl group to amino group
(b) Amino group to carboxyl group
(c) R group to carboxyl group
(d) Water to amino group

Answer: (a) Carboxyl group to amino group — It is an internal salt.

106. The acidic part of an amino acid molecule is actually the:

(a) –COOH group
(b) $-NH_2$ group
(c) $-COO^-$ group
(d) $-NH_3^+$ group

Answer: (d) $-NH_3^+$ group — Stated in the textbook.

107. The basic part of an amino acid molecule is actually the:

(a) $-NH_2$ group
(b) $-NH_3^+$ group
(c) $-COO^-$ group
(d) –COOH group

Answer: (c) $-COO^-$ group — Stated in the textbook.

108. At low pH an amino acid exists mainly in the:

(a) Anionic form (b) Cationic form
(c) Dipolar form (d) Neutral molecular form

Answer: (b) Cationic form — It is protonated.

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109. At high pH an amino acid exists mainly in the:

(a) Cationic form (b) Zwitterion form
(c) Dipolar form (d) Anionic form

Answer: (d) Anionic form — It loses a proton.

110. The pH at which the zwitterion concentration is maximum is the:

(a) Neutral point (b) End point
(c) Isoelectric point (d) Boiling point

Answer: (c) Isoelectric point — pI.

111. Electrophoresis separates amino acids on the basis of their:

(a) Charge (b) Colour
(c) Smell (d) Boiling point

Answer: (a) Charge — In an electric field.

112. If the pI of an amino acid is greater than the pH of the solution, it moves towards the:

(a) Anode (b) Cathode
(c) Neither electrode (d) Both electrodes

Answer: (b) Cathode — It carries a positive charge.

113. At pH 6, glutamic acid (pI = 3.22) moves towards the:

(a) Anode (b) Cathode
(c) Centre (d) Top

Answer: (a) Anode — It carries a negative charge.

114. The locating agent for amino acids is:

(a) Iodine (b) Ninhydrin
(c) Methyl orange (d) 2,4-DNPH

Answer: (b) Ninhydrin — It gives a purple product.

115. Ninhydrin with amino acids gives a product which is:

(a) Yellow (b) Orange
(c) Purple (d) Green

Answer: (c) Purple — Stated in the textbook.

116. Electrophoresis is only:

(a) A large-scale separation method
(b) A method of hydrolysis
(c) A method of synthesis
(d) An analytical method

Answer: (d) An analytical method — It cannot separate large quantities.

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117. Large quantities of amino acids are separated by:

(a) Electrophoresis (b) Distillation
(c) Column chromatography (d) Sublimation

Answer: (c) Column chromatography — Stated in the textbook.

118. The bond joining two amino acids is called a:

(a) Glycosidic bond (b) Peptide bond
(c) Ester bond (d) Hydrogen bond

Answer: (b) Peptide bond — An amide bond, –CO–NH–.

119. The formation of a dipeptide is a:

(a) Condensation reaction (b) Addition reaction
(c) Redox reaction (d) Hydrolysis

Answer: (a) Condensation reaction — Water is eliminated.

120. The number of peptide bonds in a tripeptide is:

(a) One (b) Four
(c) Three (d) Two

Answer: (d) Two — Three amino acids are joined by two links.

121. A compound formed from a large number of amino acids is a:

(a) Dipeptide (b) Tripeptide
(c) Polypeptide (d) Zwitterion

Answer: (c) Polypeptide — Definition in the textbook.

122. Of two amino acids with similar pI, the one that moves more slowly has the:

(a) Larger molecular weight
(b) Smaller molecular weight
(c) Higher charge
(d) Lower pI

Answer: (a) Larger molecular weight — The charge has to carry a greater mass.

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